



Deep Learning

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https://github.com/safayani/deep_learning_course



The slides are modified, based on original slides by [Geoffrey Hinton, Toronto university], [Andrew NG, Stanford university] and [Fei Fei Lee, Justin Johnson and Serena Yeung, standard university].

Why do we need to neural networks?

What is Machine Learning?

- It is very hard to write programs that solve problems like recognizing a three-dimensional object from a novel viewpoint in new lighting conditions in a cluttered scene.
 - We don't know what program to write because we don't know how its done in our brain.
 - Even if we had a good idea about how to do it, the program might be horrendously complicated.
- It is hard to write a program to compute the probability that a credit card transaction is fraudulent.
 - There may not be any rules that are both simple and reliable. We need to combine a very large number of weak rules.
 - Fraud is a moving target. The program needs to keep changing.

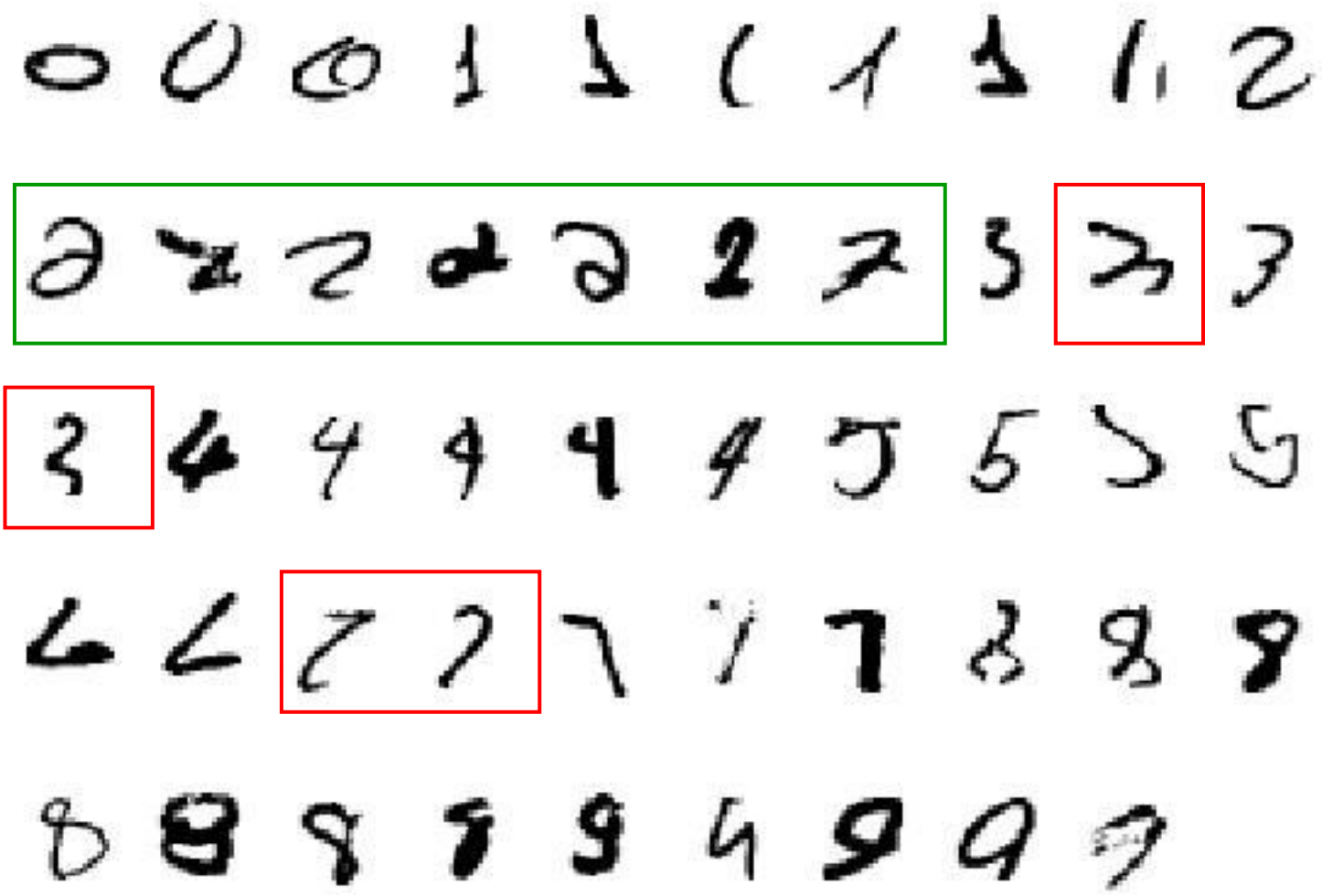
The Machine Learning Approach

- Instead of writing a program by hand for each specific task, we collect lots of examples that specify the correct output for a given input.
- A machine learning algorithm then takes these examples and produces a program that does the job.
 - The program produced by the learning algorithm may look very different from a typical hand-written program. It may contain millions of numbers.
 - If we do it right, the program works for new cases as well as the ones we trained it on.
 - If the data changes the program can change too by training on the new data.
- Massive amounts of computation are now cheaper than paying someone to write a task-specific program.

A standard example of machine learning

- The MNIST database of hand-written digits is the the machine learning equivalent of fruit flies.
 - They are publicly available and we can learn them quite fast in a moderate-sized neural net.
 - We know a huge amount about how well various machine learning methods do on MNIST.
- We will use MNIST as our standard task.

It is very hard to say what makes a 2

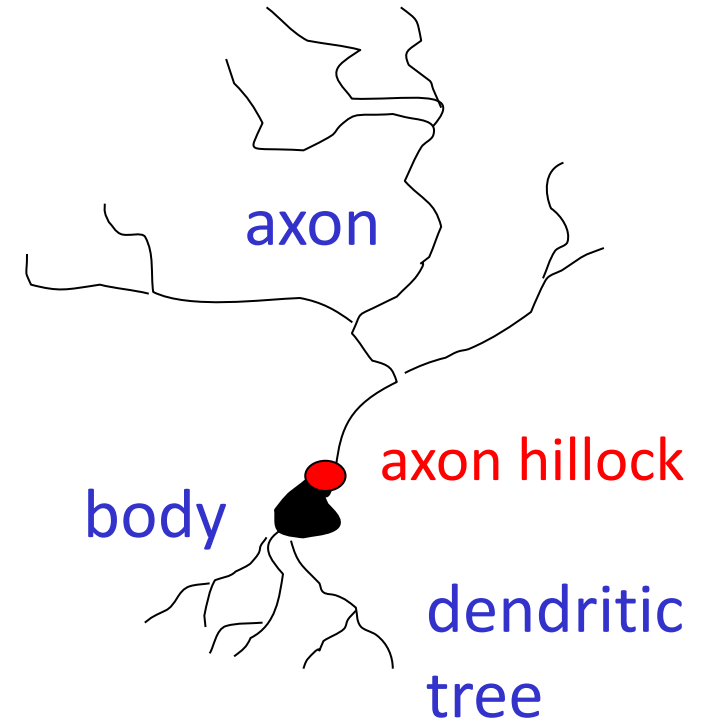


Neural Networks

What are neural networks?

A typical cortical neuron

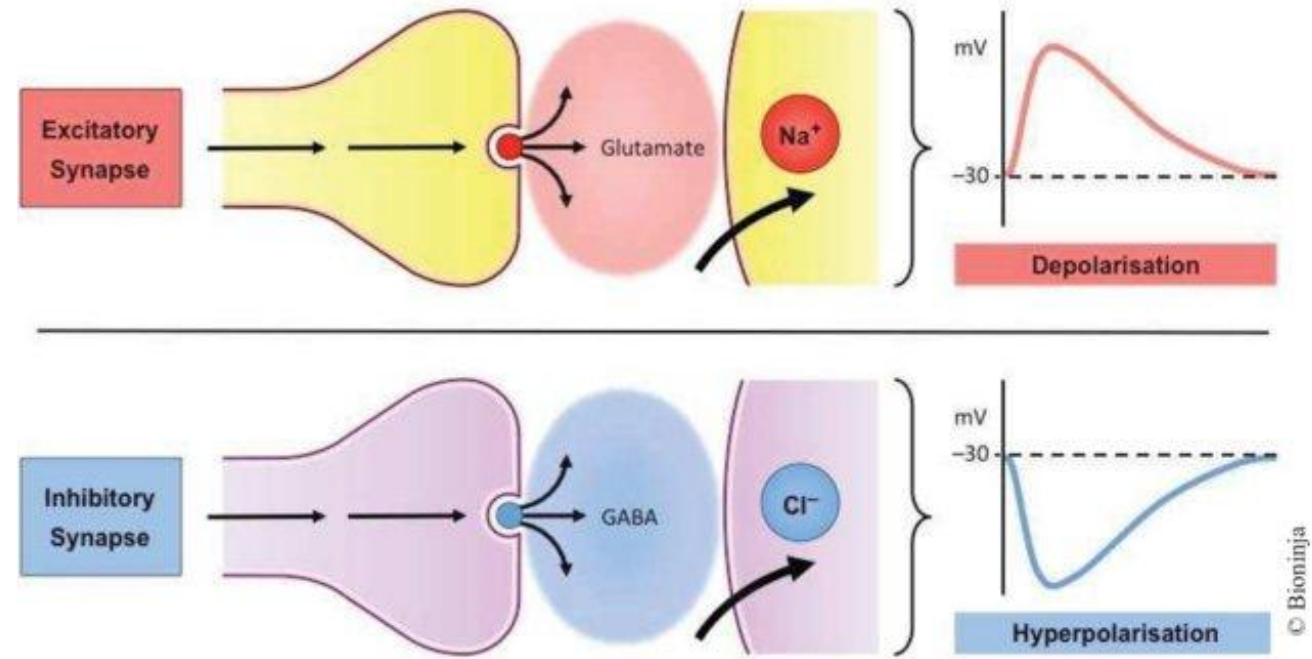
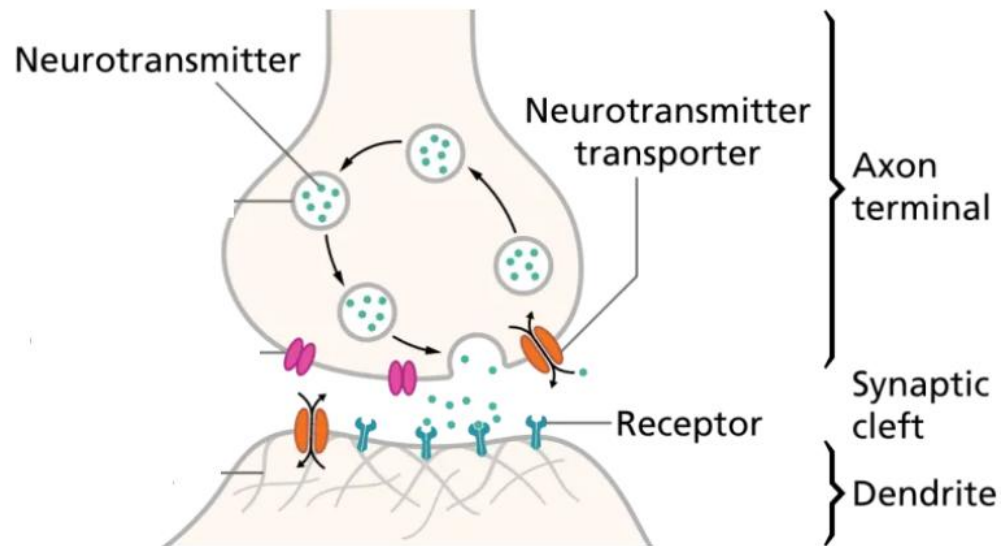
- Gross physical structure:
 - There is one axon that branches
 - There is a dendritic tree that collects input from other neurons.
- Axons typically contact dendritic trees at synapses
 - A spike of activity in the axon causes charge to be injected into the post-synaptic neuron.
- Spike generation:
 - There is an **axon hillock** that generates outgoing spikes whenever enough charge has flowed in at synapses to depolarize the cell membrane.



Synapses

- When a spike of activity travels along an axon and arrives at a synapse it causes vesicles of transmitter chemical to be released.
 - There are several kinds of transmitter.
- The transmitter molecules diffuse across the synaptic cleft and bind to receptor molecules in the membrane of the post-synaptic neuron thus changing their shape.
 - This opens up holes that allow specific ions in or out.

Synapses

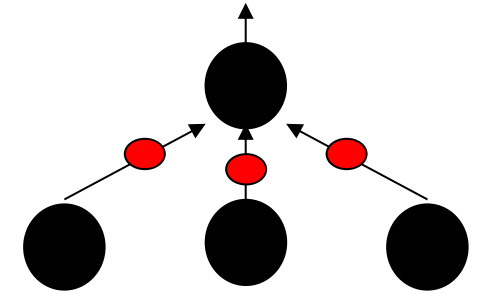


How synapses adapt

- The effectiveness of the synapse can be changed:
 - vary the number of vesicles of transmitter.
 - vary the number of receptor molecules.
- Synapses are slow, but they have advantages over RAM
 - They are very small and very low-power.
 - They adapt using locally available signals
 - But what rules do they use to decide how to change?

How the brain works on one slide!

- Each neuron receives inputs from other neurons
 - A few neurons also connect to receptors.
 - Cortical neurons use spikes to communicate.
- The effect of each input line on the neuron is controlled by a synaptic weight
 - The weights can be positive or negative.
- The synaptic weights **adapt** so that the whole network learns to perform useful computations
 - Recognizing objects, understanding language, making plans, controlling the body.
- You have about 10^{11} neurons each with about 10^4 weights.
 - A huge number of weights can affect the computation in a very short time. Much better bandwidth than a workstation.



Modularity and the brain

- Different bits of the cortex do different things.
 - Local damage to the brain has specific effects.
 - Specific tasks increase the blood flow to specific regions.
- But cortex looks pretty much the same all over.
 - Early brain damage makes functions relocate.
- Cortex is made of general purpose stuff that has the ability to turn into special purpose hardware in response to experience.
 - This gives rapid parallel computation plus flexibility.
 - Conventional computers get flexibility by having stored sequential programs, but this requires very fast central processors to perform long sequential computations.

Object Recognition

PASCAL Visual Object Challenge (20 object categories)

[Everingham et al. 2006-2012]

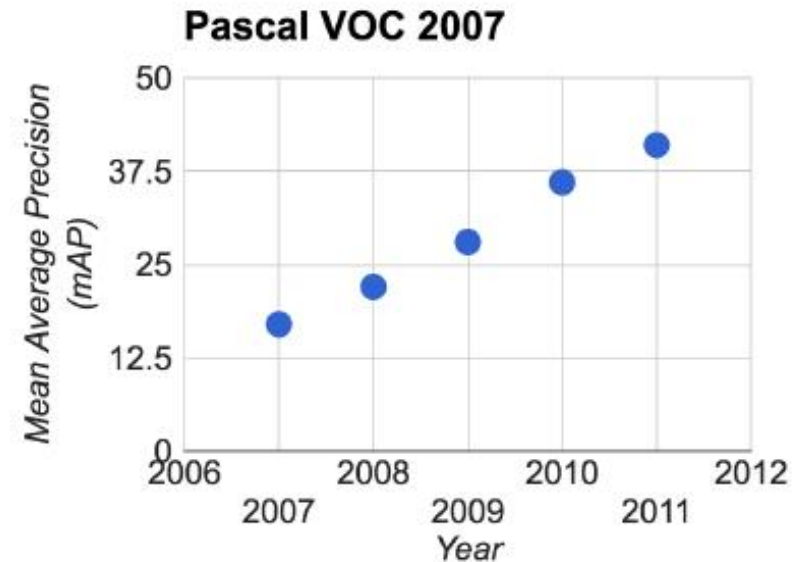
Image is CC0 1.0 public domain



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www.image-net.org

22K categories and **14M** images

- Animals
 - Bird
 - Fish
 - Mammal
 - Invertebrate
- Plants
 - Tree
 - Flower
- Food
- Materials
- Structures
 - Artifact
 - Tools
 - Appliances
 - Structures
- Person
- Scenes
 - Indoor
 - Geological Formations
- Sport Activities

IMAGENET Large Scale Visual Recognition Challenge

The Image Classification Challenge:
1,000 object classes
1,431,167 images



Output:
Scale
T-shirt
Steel drum
Drumstick
Mud turtle



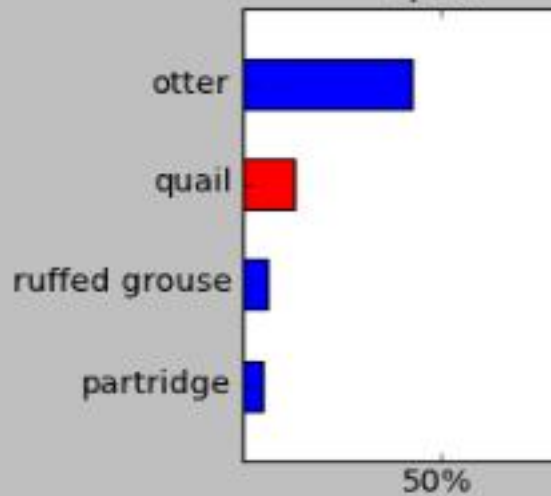
Output:
Scale
T-shirt
Giant panda
Drumstick
Mud turtle



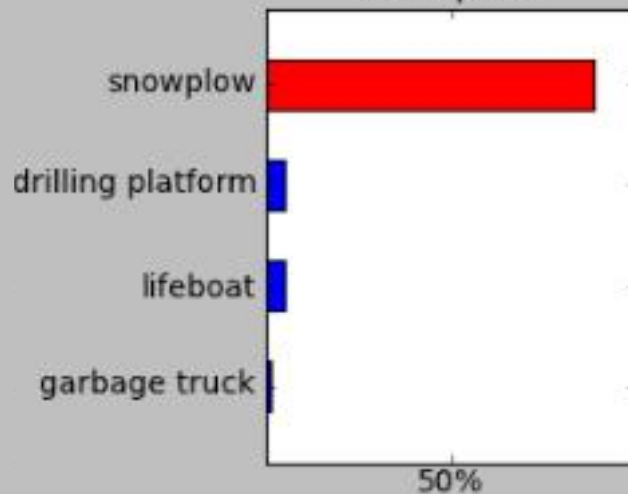
Some examples from an earlier version of the net



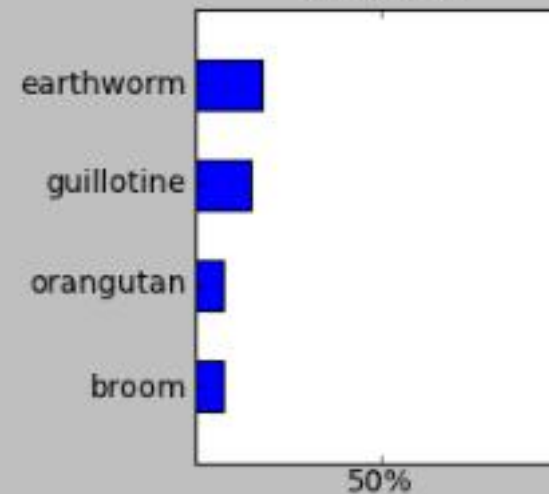
quail



snowplow



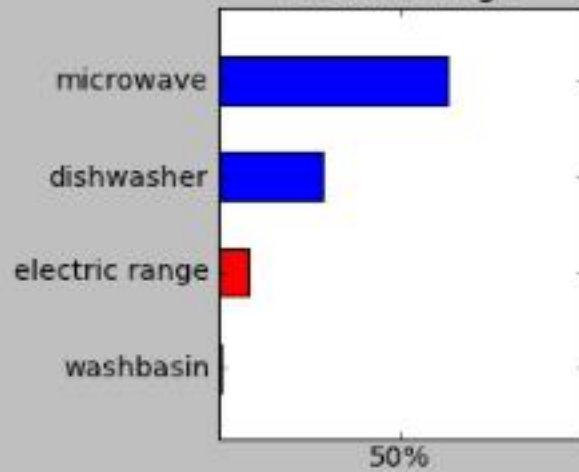
scabbard



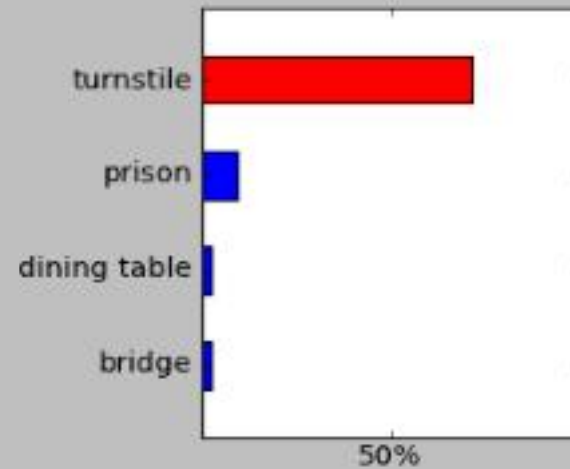
It can deal with a wide range of objects



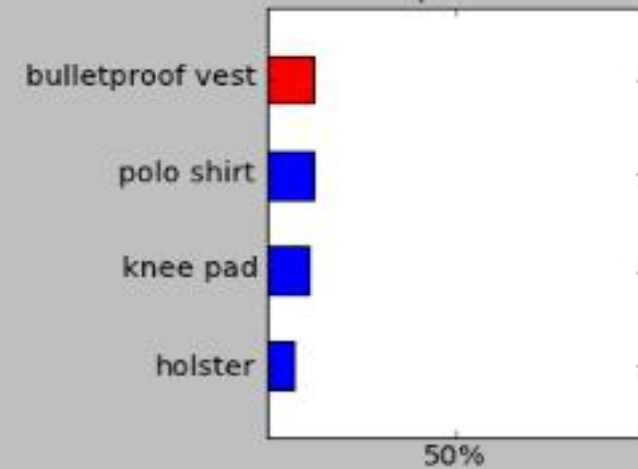
electric range



turnstile

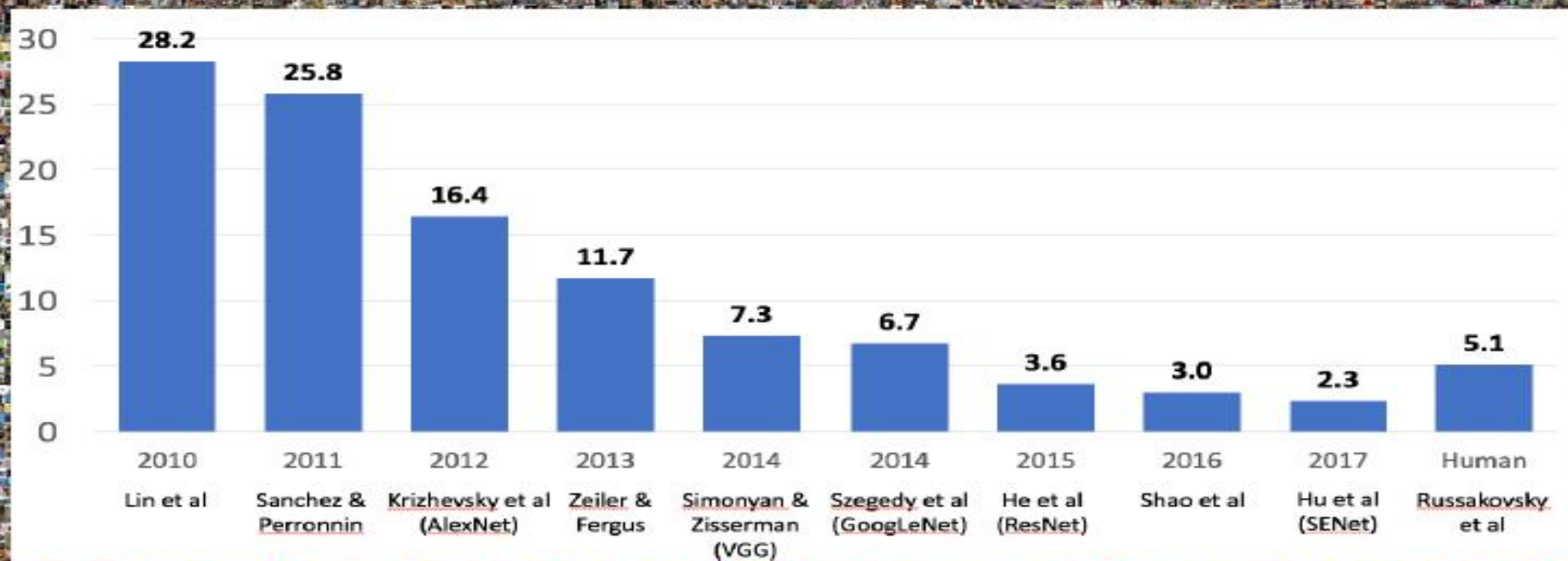


bulletproof vest

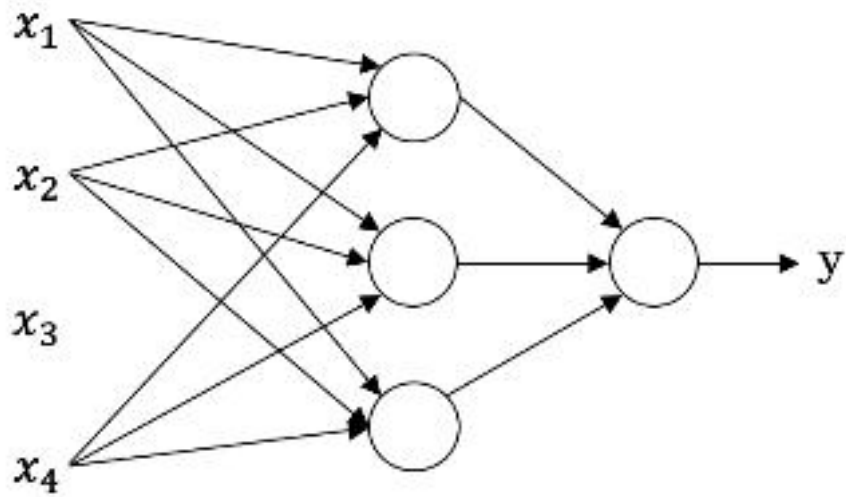


IMAGENET Large Scale Visual Recognition Challenge

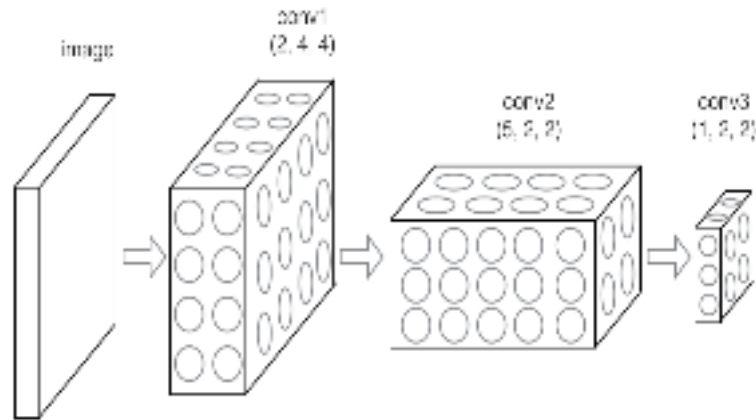
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1,431,167 images



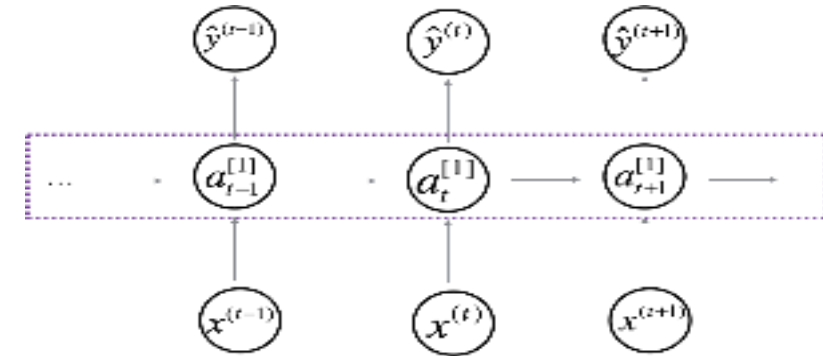
Neural Network examples



Standard NN



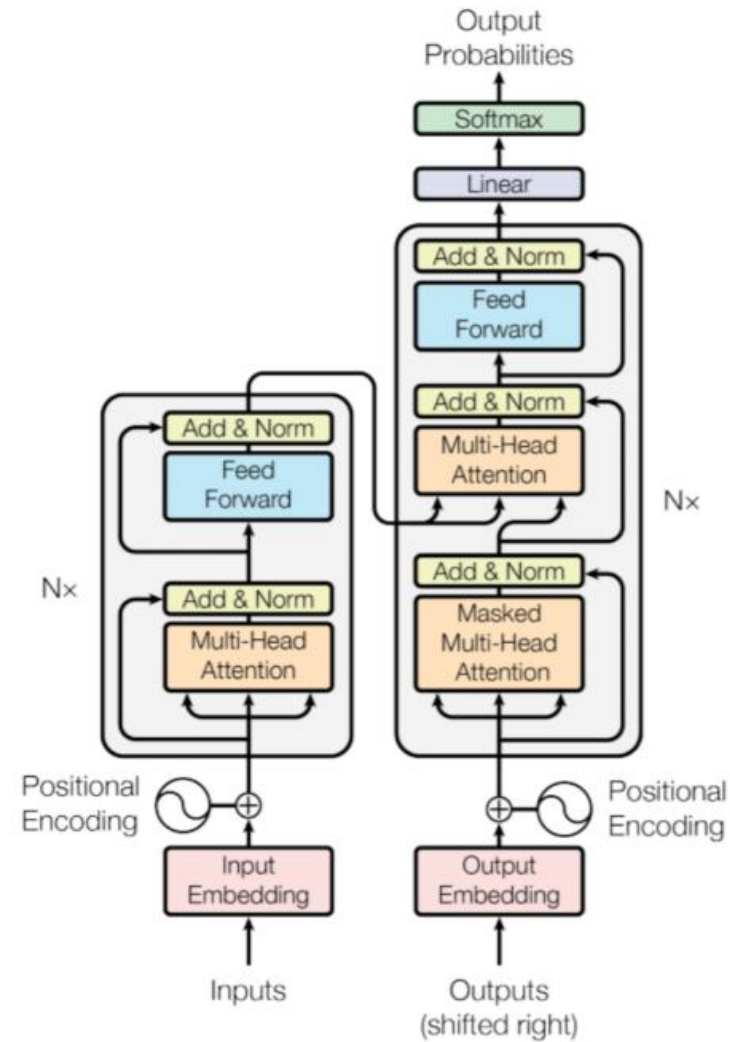
Convolutional NN



Recurrent NN

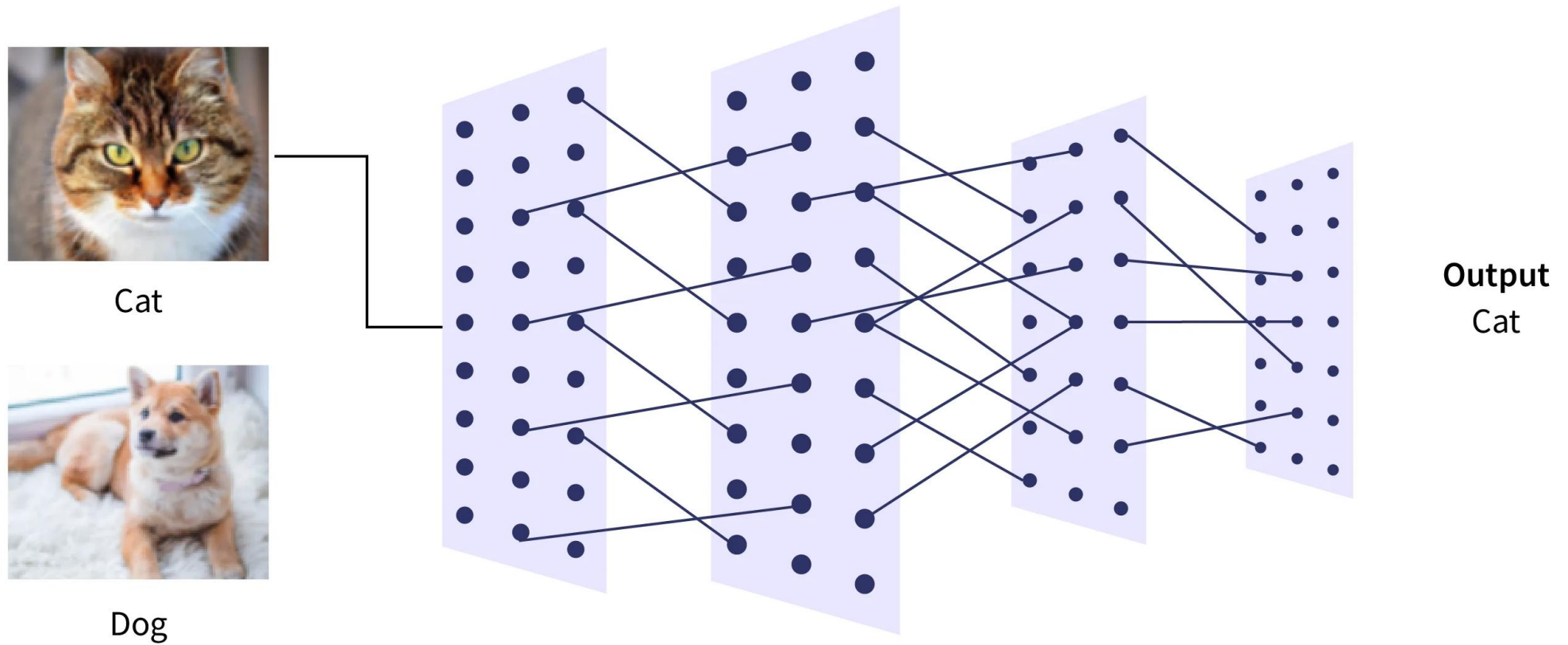
Neural Network examples

Transformers

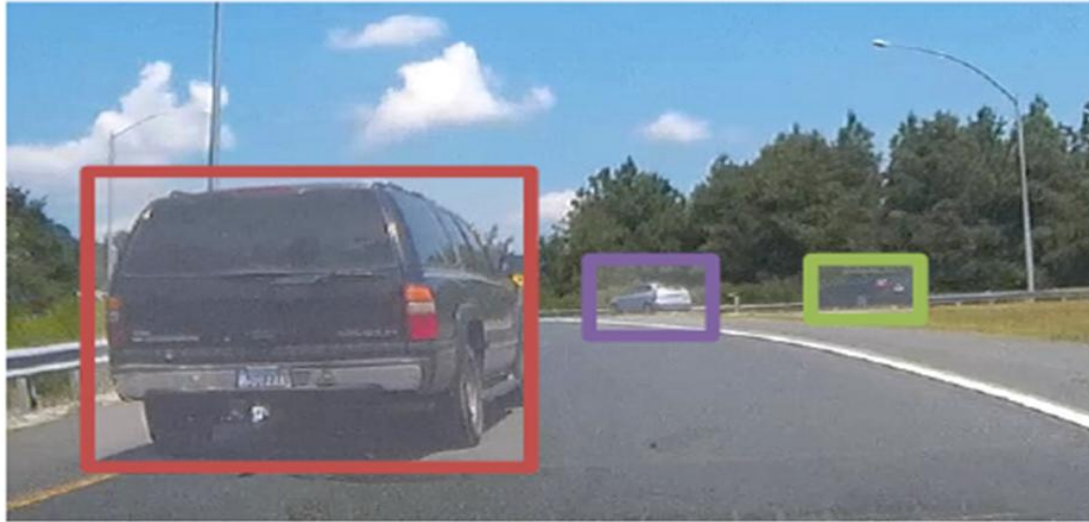


Some applications of Deep Learning

Object Classification



Object Detection



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Person

Hammer

Neural style transfer



Content



Style



Generated image



Content



Style



Generated image

Image Captioning: Example Results

Captions generated using [neuraltalk2](#).
All images are [CC0 Public domain](#):
[cat suitcase](#), [cat tree](#), [dog](#), [bear](#),
[surfers](#), [tennis](#), [giraffe](#), [motorcycle](#).



A cat sitting on a suitcase on the floor



A cat is sitting on a tree branch



A dog is running in the grass with a frisbee



A white teddy bear sitting in the grass



Two people walking on the beach with surfboards



A tennis player in action on the court



Two giraffes standing in a grassy field



A man riding a dirt bike on a dirt track

Visual Question Answering



Q: What endangered animal is featured on the truck?

A: A bald eagle.

A: A sparrow.

A: A humming bird.

A: A raven.



Q: Where will the driver go if turning right?

A: Onto 24 1/4 Rd.

A: Onto 25 3/4 Rd.

A: Onto 23 3/4 Rd.

A: Onto Main Street.



Q: Who is under the umbrella?

A: Two women.

A: A child.

A: An old man.

A: A husband and a wife.

Natural Language Processing

- Word Embedding
- Sentiment Classification
- Machine Translation
- Question answering, chat-gpt

Featurized representation: word embedding

	Man (5391)	Woman (9853)	King (4914)	Queen (7157)	Apple (456)	Orange (6257)
			-0.95	0.97	0.00	0.01
			0.93	0.95	-0.01	0.00
			0.7	0.69	0.03	-0.02
			0.02	0.01	0.95	0.97

I want a glass of orange _____.

I want a glass of apple_____.

Sentiment classification problem

x

The dessert is excellent.

Service was quite slow.

Good for a quick meal, but nothing special.

Completely lacking in good taste, good service, and good ambience.

y



Generative Models

- Generative Adversarial Network
- Variational Autoencoder

Image-to-Image Translation

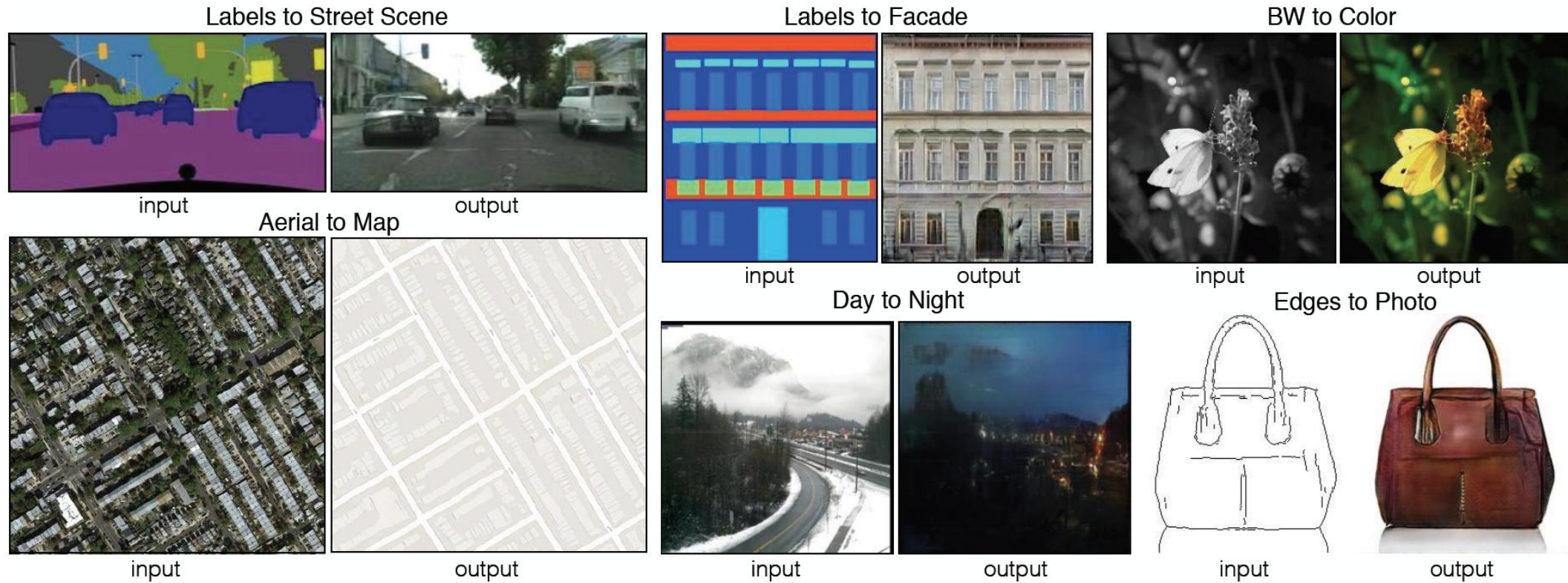


Figure 1 in the original paper.

[Link to an interactive demo of this paper](#)

Text-to-Image Synthesis

Motivation

Given a text description, generate images closely associated.

Uses a conditional GAN with the generator and discriminator being condition on “dense” text embedding.

this small bird has a pink breast and crown, and black primaries and secondaries.



this magnificent fellow is almost all black with a red crest, and white cheek patch.



the flower has petals that are bright pinkish purple with white stigma



this white and yellow flower have thin white petals and a round yellow stamen



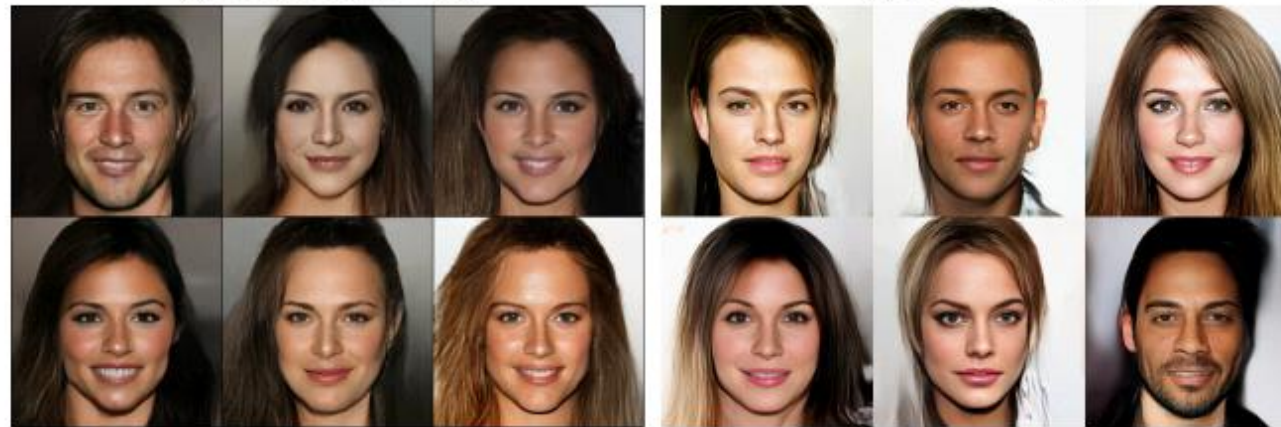
Figure 1 in the original paper.

Face images generated with a Variational Autoencoder



(d) CelebA HQ ($t = 0.6$)

(e) FFHQ ($t = 0.5$)



A. Vahdat and J. Kautz, “NVAE: A deep hierarchical variational autoencoder,” in Proc. Conf. Neural Inf. Process. Syst., 2020, pp. 19667–19679

Course outline

- Introduction
- Logistic regression
- Multi-layer perceptron (MLP)
- Back Propagation (BP)
- Regularization concept
- Optimization
- Convolution NNs
- Introducing some famous Convolution NN architectures (LeNet-5, AlexNet, VGG, google net, Resnet, mobile net, efficient net)
- Some applications of deep learning(localization, detection, face recognition, style transfer)
- Recurrent neural network (GRU, LSTM)
- Transformers
- Basic concepts of NLP
- Generative AI (GAN, VAE)

Related Courses

- Deep Learning, Andrew NG.



- Convolutional Neural Networks for Visual Recognition, Fei-Fei Lee, Justin Johnson, Serena Yeung



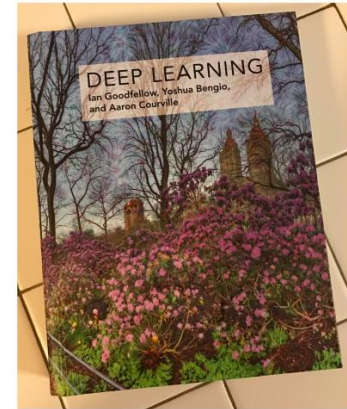
- Neural Network for Machine Learning



Textbooks

- Deep learning, I. Goodfellow, J. Bengio

MIT Press, 2016



- Neural Networks and deep learning, M. Nielsen

Determination Press, 2015



Course Strategy

- Assignments and project 5-7 marks (python)
- Midterm and quizzes 5-7 marks
- Final 5-7 marks

۶۷.۵ هزار
بازدید کل

۶۳۳
دنبال کننده

تنظیمات

هوش مصنوعی برای همه

خانه

ویدیوها

ویدیوهای کوتاه

لیست های پخش

درباره کانال

🔍

پیش بینی در بازار ارز دیجیتال با بهره گیری از مدل های زبانی بزرگ

جلسه دفاع از پایان نامه کارشناسی ارشد دانشجو: مریم فخاری استاد راهنما: دکتر مهران صفایانی مهرماه ۱۴۰۴ دانشکده مهندسی برق و کامپیوتر دانشگاه صنعتی اصفهان

۱۰۹ بازدید - ۲ ماه پیش



دانشگاه صنعتی اصفهان
دانشکده مهندسی برق و کامپیوتر

دانشجو:
مریم فخاری

استاد راهنما:
آقای دکتر مهران صفایانی

مهر ۱۴۰۴

۳۰:۰۰



سمینار دفاع از پایان نامه کارشناسی ارشد گرایش نرم افزار

**پیش بینی سری های زمانی مالی
در بازار ارز دیجیتال
با بهره گیری از مدل های زبانی بزرگ**